

Exploring new physics with proton interactions at the LHC



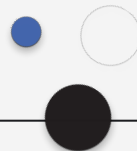
LABORATÓRIO DE INSTRUMENTAÇÃO
E FÍSICA EXPERIMENTAL DE PARTÍCULAS

Work done so far

Central exclusive production processes in proton-proton collisions.

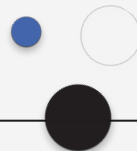
Recent research has focused on lepton pair, WW/ZZ boson and top quark ($t\bar{t}$) production, leading to:

- Precision Tests of the Standard Model
- Enhanced Sensitivity to New Physics: the ability to isolate exclusive events makes CEPs valuable for searching BSM;
- Forward Proton Tagging: the use of forward proton detectors – like PPS – improves event selection and provides more information about scattered protons, leading to more accurate measurements;



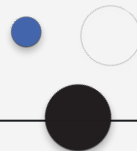
Why proton–proton collisions?

- Proton collisions at the LHC offer a unique environment for probing the Standard Model (SM)
- These collisions allow exploration of rare processes and potential new physics that are not accessible in lower-energy experiments.



Future work?

- Exploring the Anomalous Magnetic Moment of the Tau Lepton: unlike electrons and muons, leptons have higher mass and shorter lifetime, making it challenging to study the tau's $g-2$ - requires high-energy experiments;
- Since taus are 3rd generation of leptons and heavier than electrons and muons, they may be even more sensitive to Beyond SM physics;



Thank you
for your attention